

Algebra I Inequalities Cheatsheet

Module 3

Inequality Symbols

$$\begin{array}{ccccccc} & < & > & \leq & \geq & & \\ \text{open circle:} & < & \text{or} & > & & \text{closed circle:} & \leq & \text{or} & \geq \end{array}$$

Solving Rule

Solve like an equation, except flip the inequality sign when multiplying or dividing by a negative number.

$$-3x < 12 \Rightarrow x > -4$$

Interval Notation

$$x < 5 \Rightarrow (-\infty, 5)$$

$$x \geq -2 \Rightarrow [-2, \infty)$$

$$-1 < x \leq 4 \Rightarrow (-1, 4]$$

Parentheses exclude endpoints. Brackets include endpoints.

Compound Inequalities

AND means overlap:

$$x > 2 \text{ and } x < 7 \Rightarrow 2 < x < 7$$

OR means union:

$$x < -3 \text{ or } x \geq 5$$

Three-Part Inequalities

$$-4 < 2x + 6 \leq 10$$

Subtract 6 from all three parts:

$$-10 < 2x \leq 4$$

Divide all three parts by 2:

$$-5 < x \leq 2$$

Absolute Value Inequalities

$$|A| < c \Rightarrow -c < A < c \quad (c > 0)$$

$$|A| \leq c \Rightarrow -c \leq A \leq c \quad (c \geq 0)$$

$$|A| > c \Rightarrow A < -c \text{ or } A > c \quad (c \geq 0)$$

$$|A| \geq c \Rightarrow A \leq -c \text{ or } A \geq c \quad (c \geq 0)$$

Quick Checks

- Flip the inequality only when multiplying or dividing by a negative.
- Use AND for values between two endpoints.
- Use OR for values outside two endpoints.
- No number has negative absolute value.